

Chapter 1

Introduction of Microprocessor

A Microprocessor is a multipurpose programmable, clock driven, register based electronic device that reads binary instructions from a storage device called memory, accepts binary data as input, processes data according to those instructions and provide results as output. The microprocessor operates in binary 0 and 1 known as bits are represented in terms of electrical voltages in the machine that means 0 represents low voltage level and 1 represents high voltage level. Each microprocessor recognizes and processes a group of bits called the word and microprocessors are classified according to their word length such as 8 bits microprocessor with 8 bit word and 32 bit microprocessor with 32 bit word etc.

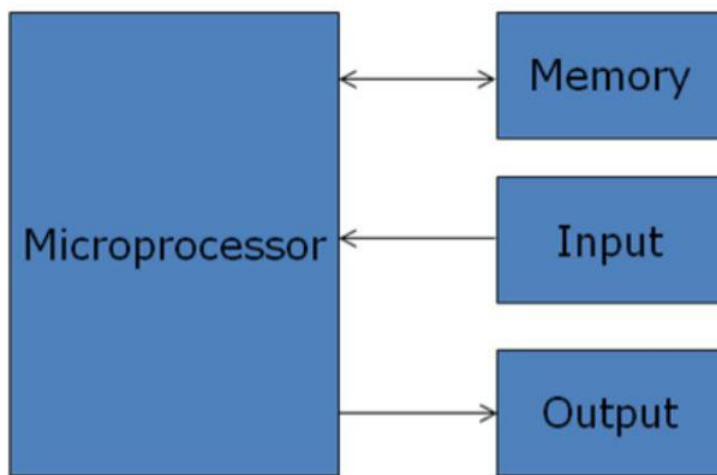


Fig : Block diagram of programmable machine/device

A machine consists of physical components such as microprocessor memory. Input and output unit and they are called hardware equipment/components. A set of instructions return further for the microprocessor to perform a task is called a program. A group of program is called software. Some examples of Microprocessor are 8008, 8080, 8085, 8086 etc.

Evolution of Microprocessor

Generations of microprocessor:

1. First generation

From 1971 to 1972 the era of the first generation came which brought microprocessors like INTEL 4004 Rockwell international PPS-4 INTEL 8008 etc.

2. Second generation

The second generation marked the development of 8 bit microprocessors from 1973 to 1978. Processors like INTEL 8085 Motorola 6800 and 6801 etc came into existence.

3. Third generation

The third generation brought forward the 16 bit processors like INTEL 8086/80186/80286 Motorola 68000 68010 etc. From 1979 to 1980 this generation used the HMOS technology.

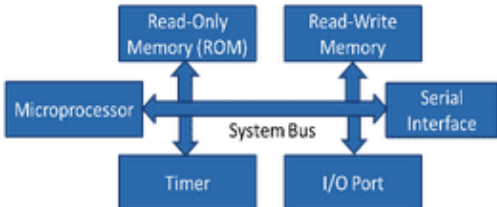
4. Fourth generation

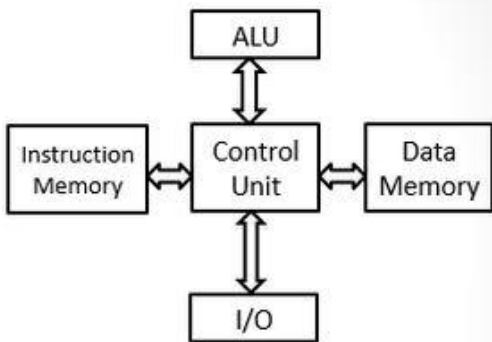
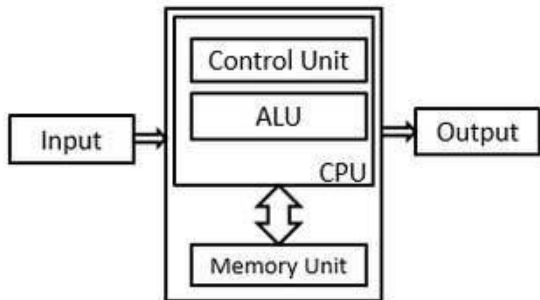
The fourth generation came into existence from 1981 to 1995. The 32 bit processors using HMOS fabrication came into existence. INTEL 80386 and Motorola 68020 are some of the popular processors of this generation.

5. Fifth generation

From 1995 till now we are in the fifth generation. 64 bit processors like PENTIUM, celeron, dual, quad and octa core processors came into existence.

Processor Name	Introduced Date	Transistors	Microns	Clock Speed	Size of Data	MIPS
8080	1974	6000	6	2 MHz	8 bits	0.64
8088	1979	29000	3	5 MHz	16 bits 8-bit bus	0.33
80286	1982	134,000	1.5	6 MHz	16 bits	1
80386	1985	275,000	1.5	16 MHz	32 bits	5
80486	1989	1,200,000	1	25 MHz	32 bits	20
Pentium	1993	3,100,000	0.8	60 MHz	32 bits 64-bit bus	100
Pentium II	1997	7,500,000	0.35	233 MHz	32 bits 64-bit bus	~300
Pentium III	1999	9,500,000	0.25	450 MHz	32 bits 64-bit bus	~510
Pentium 4	2000	42,000,000	0.18	1.5 GHz	32 bits 64-bit bus	~1700
Pentium 4 "Prescott"	2004	125,000,000	0.09	3.6GHz	32 bits 64-bit bus	~7000

Microprocessor	Micro Controller
	
Microprocessor is heart of Computer system.	Micro Controller is a heart of embedded system.
It is just a processor. Memory and I/O components have to be connected externally	Micro controller has external processor along with internal memory and i/O components
Since memory and I/O has to be connected externally, the circuit becomes large.	Since memory and I/O are present internally, the circuit is small.
Cannot be used in compact systems and hence inefficient	Can be used in compact systems and hence it is an efficient technique
Cost of the entire system increases	Cost of the entire system is low
Due to external components, the entire power consumption is high. Hence it is not suitable to use with devices running on stored power like batteries.	Since external components are low, total power consumption is less and can be used with devices running on stored power like batteries.
Most of the microprocessors do not have power saving features.	Most of the micro controllers have power saving modes like idle mode and power saving mode. This helps to reduce power consumption even further.
Since memory and I/O components are all external, each instruction will need external operation, hence it is relatively slower.	Since components are internal, most of the operations are internal instruction, hence speed is fast.
Microprocessor have less number of registers, hence more operations are memory based.	Micro controller have more number of registers, hence the programs are easier to write.
Microprocessors are based on von Neumann model/architecture where program and data are stored in same memory module	Micro controllers are based on Harvard architecture where program memory and Data memory are separate
Mainly used in personal computers	Used mainly in washing machine, MP3 players

Point of Comparison	Harvard Architecture	Von Neumann Architecture
Arrangement	In Harvard architecture, the CPU is connected with both the data memory (RAM) and program memory (ROM), separately.  Harvard Model	In Von-Neumann architecture, there is no separate data and program memory. Instead, a single memory connection is given to the CPU.  Von Neumann Model
Hardware requirements	It requires more hardware since it will be requiring separate data and address bus for each memory.	In contrast to the Harvard architecture, this requires less hardware since only a common memory needs to be reached.
Space requirements	This requires more space.	Von-Neumann Architecture requires less space.
Speed of execution	Speed of execution is faster because the processor fetches data and instructions simultaneously .	Speed of execution is slower since it cannot fetch the data and instructions at the same time.
Space usage	It results in wastage of space since if the space is left in the data memory then the instructions memory cannot use the space of the data memory and vice-versa.	Space is not wasted because the space of the data memory can be utilized by the instructions memory and vice-versa.
Controlling	Controlling becomes complex since data and instructions are to be fetched simultaneously.	Controlling becomes simpler since either data or instructions are to be fetched at a time.

Difference between Harvard and Von-Neumann Architecture

Difference between CISC Vs RISC

CISC	RISC
The original microprocessor ISA	Redesigned ISA that emerged in the early 1980s
Instructions can take several clock cycles	Single-cycle instructions
Hardware-centric design – the ISA does as much as possible using hardware circuitry	Software-centric design – High-level compilers take on most of the burden of coding many software steps from the programmer
More efficient use of RAM than RISC	Heavy use of RAM (can cause bottlenecks if RAM is limited)
Complex and variable length instructions	Simple, standardized instructions
May support microcode (micro-programming where instructions are treated like small programs)	Only one layer of instructions
Large number of instructions	Small number of fixed-length instructions
Compound addressing modes	Limited addressing modes

Internal Architecture of 8085 microprocessor and its registers

The Intel 8085 A is a complete 8 bit parallel central processing unit. The main components of 8085A are array of registers, the arithmetic logic unit, the encoder/decoder, and timing and control circuits linked by an internal data bus. The block diagram is shown below:

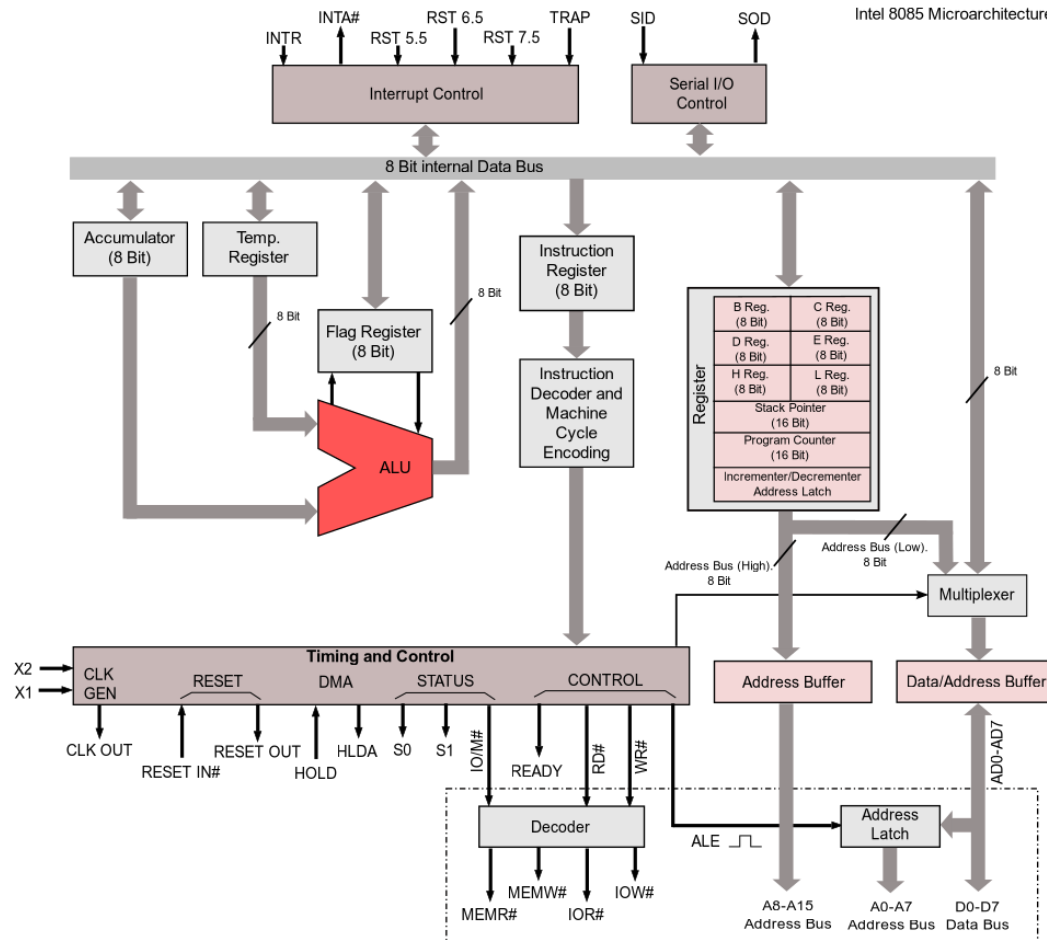


Fig: The 8085 A microprocessor functional block diagram

1. ALU

The arithmetic logic unit performs the computing functions, it includes the accumulator, the temporary register, the arithmetic and logic circuits and five flags. The temporary register is used to hold data during an arithmetic/logic operation. The result is stored in the accumulator; the flags (flip-flops) are set or reset according to the result of the operation.

2. Accumulator (register A)

It is an 8 bit register that is the part of ALU. This register is used to store the 8-bit data and to perform arithmetic and logic operations and 8085 microprocessor is called accumulator based microprocessor. When data is read from input port, it first moved to accumulator and when data is sent to output port, it must be first placed in accumulator.

3. Temporary registers(W & Z)

They are 8 bit registers not accessible to the programmer. During program execution, 8085A places the data into it for a brief period.

4. Instruction register(IR)

It is an 8 bit register not accessible to the programmer. It receives the operation codes of instruction from internal data bus and passes to the instruction decoder which decodes so that microprocessor knows which type of operation is to be performed.

5. Register Array

(Scratch pad registers B, C, D, E): It is a 8 bit register accessible to the programmers. Data can be stored upon it during program execution. These can be used individually as 8-bit registers or in pair BC, DE as 16 bit registers. The data can be directly added or transferred from one to another. Their contents may be incremented or decremented and combined logically with the content of the accumulator.

a. Register H & L

They are 8 bit registers that can be used in same manner as scratch pad registers.

b. Stack Pointer (SP)

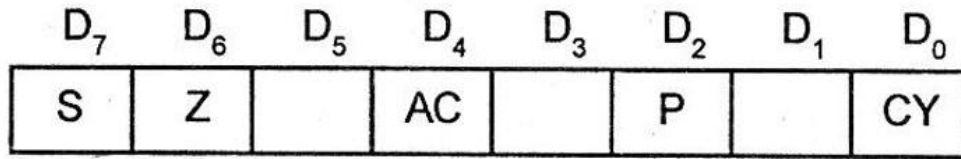
It is a 16 bit register used as a memory pointer. It points to a memory location in R/W memory, called the stack. The beginning of the stack is defined by loading a 16- bit address in the stack pointer.

c. Program Counter (PC)

Microprocessor uses the PC register to sequence the execution of the instructions. The function of PC is to point to the memory address from which the next byte is to be fetched. When a byte is being fetched, the PC is incremented by one to point to the next memory location.

6. Flags

FLAG REGISTER OF 8085



Flag is an 8-bit register containing 5 1-bit flags:

Sign - set if the most significant bit of the result is set.

Zero - set if the result is zero.

Auxiliary carry - set if there was a carry out from bit 3 to bit 4 of the result.

Parity - set if the parity (the number of set bits in the result) is even.

Carry - set if there was a carry during addition, or borrow during subtraction/comparison.

Register consists of five flip flops, each holding the status of different states separately is known as flag register and each flip flop are called flags. 8085A can set or reset one or more of the flags and are sign(S), Zero (Z), Auxiliary Carry (AC) and Parity (P) and Carry (CY). The state of flags indicates the result of arithmetic and logical operations, which in turn can be used for decision making processes. The different flags are described as:

- **Carry**
If the last operation generates a carry its status will be 1 otherwise 0. It can handle the carry or borrow from one word to another.
- **Zero**
If the result of last operation is zero, its status will be 1 otherwise 0. It is often used in loop control and in searching for particular data value.
- **Sign**
If the most significant bit (MSB) of the result of the last operation is 1 (negative), then its status will be 1 otherwise 0.
- **Parity**
If the result of the last operation has even number of 1's (even parity), its status will be 1 otherwise 0.
- **Auxiliary carry**
If the last operation generates a carry from the lower half word (lower nibble), its status will be 1 otherwise 0. Used for performing BCD arithmetic.

7. Timing and Control Unit

This unit synchronizes all the microprocessor operations with the clock and generates the control signals necessary for communication between the microprocessor and peripherals. The control signals are similar to the sync pulse in an oscilloscope. The $\overline{RD}$ and $\overline{WR}$ signals are sync pulses indicating the availability of data on the data bus.

8. Interrupt controls

The various interrupt controls signals (INTR, RST 5.5, RST 6.5, RST 7.5 and TRAP) are used to interrupt a microprocessor.

9. Serial I/O controls

Two serial I/O control signals (SID and SOD) are used to implement the serial data transmission.

10. Address bus and data bus

Data bus carries the data to be stored. It is bidirectional, whereas address bus carries the location to where it should be stored and it is unidirectional. It is used to transfer the data & Address I/O devices.

11. Address buffer and address-data buffer

The content stored in the stack pointer and program counter is loaded into the address buffer and address-data buffer to communicate with the CPU. The memory and I/O chips are connected to these buses; the CPU can exchange the desired data with the memory and I/O chips.

Fetch-decode-execute cycle

The main job of the CPU is to execute programs using the fetch-decode-execute cycle (also known as the instruction cycle). This cycle begins as soon as you turn on a computer.

To execute a program, the program code is copied from secondary storage into the main memory. The CPU's program counter is set to the memory location where the first instruction in the program has been stored, and execution begins. The program is now running.

In a program, each machine code instruction takes up a slot in the main memory. These slots (or memory locations) each have a unique memory address. The program counter stores the address of each instruction and tells the CPU in what order they should be carried out.

When a program is being executed, the CPU performs the fetch-decode-execute cycle, which repeats over and over again until reaching the STOP instruction.

Summary of the fetch-decode-execute cycle

1. The processor checks the program counter to see which instruction to run next.
2. The program counter gives an address value in the memory of where the next instruction is.

3. The processor fetches the instruction value from this memory location.
4. Once the instruction has been fetched, it needs to be decoded and executed. For example, this could involve taking one value, putting it into the ALU, then taking a different value from a register and adding the two together.
5. Once this is complete, the processor goes back to the program counter to find the next instruction.
6. This cycle is repeated until the program ends.